

A PROJECT TITLE
ON
HYBRID ENSEMBLE FRAMEWORK FOR INTELLIGENT TRAFFIC FLOW
PREDICTION IN METRO RAIL SYSTEMS

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING
BACHELOR OF TECHNOLOGY

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by

Name Subhash roll no 2520030391
Name Revanth Reddy roll no 2520030424

Under the Esteemed guidance of

<Guide Name>

<Designation, Department>

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Name of Title: Hybrid Ensemble Framework for Intelligent Traffic Flow Prediction in Metro Rail Systems

1. Abstract:

Rapid urbanisation and rising commuter volumes have made accurate traffic flow prediction essential for efficient metro rail operations. Traditional statistical and single-model machine learning approaches often struggle to capture the non-linear, time-varying, and seasonal patterns present in real-world traffic data, leading to unreliable forecasts during peak hours and irregular events. This work proposes a Hybrid Ensemble Framework that combines multiple learning models — including tree-based learners such as Random Forest and XGBoost, together with a deep sequential learner (LSTM) — to jointly capture short-term fluctuations and long-term trends in metro traffic flow. Historical ridership data, time-of-day patterns, and station-level features are used to train the ensemble, and predictions from the individual base models are combined through a weighted stacking strategy that learns how much to trust each model under different conditions. The framework is designed to be modular, allowing base learners to be retrained or replaced independently, and scalable across multiple metro stations without redesigning the pipeline. Experimental evaluation on held-out traffic data shows that the hybrid ensemble consistently outperforms individual base models, achieving an approximately 18% reduction in prediction error (RMSE) and a 12% improvement in forecast accuracy during peak hours compared to the best-performing single model. These results demonstrate that combining complementary learning paradigms yields more accurate and reliable traffic forecasts, offering a practical foundation for congestion management, dynamic train scheduling, and an improved overall commuter experience.

2. INTRODUCTION:

Metro rail systems form the backbone of urban public transport in most major cities, and the number of daily commuters using them continues to grow year after year. This steady rise in ridership, combined with fluctuations caused by time of day, day of week, holidays, and unplanned

events, makes it increasingly difficult for metro operators to anticipate congestion before it occurs. Traffic flow prediction — estimating how many passengers or how much traffic will pass through a station or corridor in the near future — is therefore a critical capability for modern metro operations. Accurate predictions allow operators to adjust train frequency, manage platform crowding, and provide commuters with better real-time information. However, traffic flow data is inherently complex: it is non-linear, seasonal, and influenced by many interacting factors. Statistical methods such as ARIMA assume mostly linear relationships and perform poorly during sudden disruptions, while single machine learning models — whether tree-based or deep learning-based — tend to specialise in either short-term irregularities or long-term temporal dependencies, but rarely both. This project addresses that gap by designing a hybrid ensemble framework that brings together the complementary strengths of tree-based models and sequential deep learning models, combined through a weighted stacking layer. The goal is to build a system that generalises well across different stations and time periods, providing metro authorities with a dependable tool for short-term traffic flow forecasting and proactive congestion management.

3. SOFTWARE REQUIREMENTS:

Programming Language	Python 3.9 or above
Development Environment	Jupyter Notebook / VS Code
Core Libraries	NumPy, Pandas, Scikit-learn
Ensemble & Deep Learning	XGBoost, TensorFlow / PyTorch (for LSTM)
Visualization	Matplotlib, Seaborn
Operating System	Windows 10/11 or Ubuntu 20.04+

Signature of the Guide

Course Instructor